

Limits of polynomials and rational functions



Outline

Limits of polynomials at $\pm\infty$

Limits of rational functions at $\pm\infty$

Limits of quotients of sums of powers with rational exponents

Limits of polynomials at $\pm\infty$

Introduction: limits of polynomials at $\pm\infty$

- ▶ In this section we see how to compute the limit of a polynomial as $x \rightarrow \pm\infty$
- ▶ The key idea is to **factor out the monomial of maximum degree**: all the other terms become negligible
- ▶ This technique avoids indeterminate forms of the type $+\infty - \infty$ and will be the basis for limits of rational functions

Polynomials

- ▶ A polynomial of (maximum) degree n can be written as:

$$P_n(x) = \sum_{i=0}^n a_i x^i$$

with $a_i \in \mathbb{R}$, for $i \in \{0, 1, \dots, n\}$, and $a_n \neq 0$.

- ▶ The value a_i is the **coefficient of the monomial i** , with $i = 0, 1, \dots, n$, while x^i is the **literal part** of the monomial.
- ▶ In this notation, without loss of generality, the monomials are ordered by increasing values of the exponents.
- ▶ If a monomial of degree i is missing we have $a_i = 0$; a_0 is the constant term since $x^0 = 1$.

Factoring out the monomial of maximum degree

- Factoring out $a_n x^n$, that is, the last monomial, which is the one of maximum degree, we have:

$$P_n(x) = a_n x^n \left(\overbrace{\frac{a_0}{a_n x^n} + \frac{a_1}{a_n x^{n-1}} + \dots + \frac{a_{n-1}}{a_n x}}^{\rightarrow 0 \text{ as } x \rightarrow \pm \infty} + 1 \right)$$

Limit of a polynomial at $\pm\infty$

$$\lim_{x \rightarrow \pm\infty} P_n(x) = \lim_{x \rightarrow \pm\infty} a_n x^n$$

To compute the limit of a polynomial as $x \rightarrow \pm\infty$, it is enough to compute the limit of the monomial of maximum degree.

Example: limits of polynomials as $x \rightarrow \pm\infty$

For example:

$$\begin{aligned}\lim_{x \rightarrow +\infty} (8 - x^2 + 3x^3) &= \lim_{x \rightarrow +\infty} 3x^3 \left(\underbrace{\frac{8}{3x^3} - \frac{1}{3x}}_{\rightarrow 0 \text{ as } x \rightarrow +\infty} + 1 \right) \\ &= \lim_{x \rightarrow +\infty} 3x^3 = +\infty\end{aligned}$$

$$\begin{aligned}\lim_{x \rightarrow -\infty} (120 + 4x - 2x^3) &= \lim_{x \rightarrow -\infty} -2x^3 \left(\underbrace{\frac{120}{-2x^3} + \frac{4}{-2x^2}}_{\rightarrow 0 \text{ as } x \rightarrow -\infty} + 1 \right) \\ &= \lim_{x \rightarrow -\infty} -2x^3 = +\infty\end{aligned}$$

Summary

- ▶ A polynomial of degree n has the form $P_n(x) = \sum_{i=0}^n a_i x^i$ with $a_n \neq 0$
- ▶ Factoring out $a_n x^n$, all the terms in parentheses except 1 tend to 0 as $x \rightarrow \pm\infty$
- ▶ Hence $\lim_{x \rightarrow \pm\infty} P_n(x) = \lim_{x \rightarrow \pm\infty} a_n x^n$: only the monomial of maximum degree matters

Limits of rational functions at $\pm\infty$

Introduction: limits of rational functions at $\pm\infty$

- ▶ We extend the idea of the previous section to ratios of polynomials (rational functions)
- ▶ Factoring out the monomials of maximum degree in the numerator and in the denominator resolves the indeterminate forms $\frac{\infty}{\infty}$
- ▶ We will also see how to handle the case $x \rightarrow 0$, where instead the powers of minimum degree matter

Rational functions

- Let f be a rational function (ratio of polynomials):

$$f(x) = \frac{P_n(x)}{P_m(x)}$$

where $P_n(x)$ and $P_m(x)$ are polynomials of degree n and m , respectively:

$$P_n(x) = \sum_{i=0}^n a_i x^i, \quad a_i \in \mathbb{R}, \quad i = 0, 1, \dots, n, \quad a_n \neq 0,$$

$$P_m(x) = \sum_{i=0}^m b_i x^i, \quad b_i \in \mathbb{R}, \quad i = 0, 1, \dots, m, \quad b_m \neq 0.$$

Factoring out the monomials of maximum degree

- Factoring out $a_n x^n$ in the numerator and $b_m x^m$ in the denominator, that is, the monomials of maximum degree, we have:

$$\frac{P_n(x)}{P_m(x)} = \frac{a_n x^n}{b_m x^m} \frac{\left\{ \overbrace{\frac{a_0}{a_n x^n} + \frac{a_1}{a_n x^{n-1}} + \cdots + \frac{a_{n-1}}{a_n x}}^{\rightarrow 0 \text{ as } x \rightarrow \pm\infty} + 1 \right\}}{\left\{ \underbrace{\frac{b_0}{b_m x^m} + \frac{b_1}{b_m x^{m-1}} + \cdots + \frac{b_{m-1}}{b_m x}}_{\rightarrow 0 \text{ as } x \rightarrow \pm\infty} + 1 \right\}}.$$

Limit of a rational function at $\pm\infty$

$$\lim_{x \rightarrow \pm\infty} \frac{P_n(x)}{P_m(x)} = \lim_{x \rightarrow \pm\infty} \frac{a_n x^n}{b_m x^m} = \lim_{x \rightarrow \pm\infty} \frac{a_n}{b_m} x^{n-m}$$

To compute the limit of a ratio of polynomials as $x \rightarrow \pm\infty$, it is enough to compute the limit of the ratio of the monomials of maximum degree.

Example: limits of rational functions as $x \rightarrow \pm\infty$

For example:

$$\lim_{x \rightarrow -\infty} \frac{1 + 3x^3}{2 + x^2} = \lim_{x \rightarrow -\infty} \frac{3x^3}{x^2} = \lim_{x \rightarrow -\infty} 3x = -\infty$$

$$\lim_{x \rightarrow +\infty} \frac{1 + 3x^3}{1 - 10x + x^4} = \lim_{x \rightarrow +\infty} \frac{3x^3}{x^4} = \lim_{x \rightarrow +\infty} 3 \frac{1}{x} = 0$$

$$\lim_{x \rightarrow -\infty} \frac{7x^2 + 6x^5}{-4 + 7x^5} = \lim_{x \rightarrow -\infty} \frac{6x^5}{7x^5} = \lim_{x \rightarrow -\infty} \frac{6}{7} = \frac{6}{7}$$

Limits of rational functions as $x \rightarrow 0$

To compute the limit of rational functions (without the constant term) as $x \rightarrow 0$ we must factor out, in the numerator and in the denominator, the powers of minimum degree.

Example: limits of rational functions as $x \rightarrow 0$

For example:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{-2x^2 + 3x^3 - 6x^5}{x + x^2 - x^7} &= \lim_{x \rightarrow 0} \frac{-2x^2 \left(1 - \overbrace{\frac{3}{2}x + 3x^3}^{\rightarrow 0 \text{ as } x \rightarrow 0} \right)}{x \left(1 + \underbrace{x - x^6}_{\rightarrow 0 \text{ as } x \rightarrow 0} \right)} \\ &= \lim_{x \rightarrow 0} -2x = 0\end{aligned}$$

Summary

- ▶ As $x \rightarrow \pm\infty$ a rational function behaves like the ratio of the monomials of maximum degree:

$$\lim_{x \rightarrow \pm\infty} \frac{P_n(x)}{P_m(x)} = \lim_{x \rightarrow \pm\infty} \frac{a_n}{b_m} x^{n-m}$$

- ▶ The result depends on $n - m$: $\pm\infty$ if $n > m$, $\frac{a_n}{b_m}$ if $n = m$, 0 if $n < m$
- ▶ As $x \rightarrow 0$ (without constant term) we instead factor out the powers of minimum degree

Limits of quotients of sums of powers with rational exponents



Introduction: powers with rational exponents

- ▶ The techniques seen for polynomials and rational functions extend to sums of powers with rational exponents (for example $x^{1/3}$, $x^{2/3}$)
- ▶ We will see when it is possible to consider $x \rightarrow 0$ and not only $x \rightarrow 0^+$
- ▶ Finally we introduce a second method: the **change of variable** $x = \frac{1}{y}$, which reduces limits as $x \rightarrow 0$ to limits at $\pm\infty$

The same rules

- ▶ The same rules hold for quotients of sums of powers with rational exponents:
 - as $x \rightarrow \infty$ we must factor out the power with the maximum exponent
 - as $x \rightarrow 0^+$ we must factor out the power with the minimum exponent

Powers with negative base

Raising to a power can also be defined with a negative base if the exponent is a rational number (fraction) with an odd denominator.

Hence in this case we can compute the limit as $x \rightarrow 0$ (otherwise only as $x \rightarrow 0^+$).

Example: powers with rational exponents (1/2)

For example:

$$\lim_{x \rightarrow 0} \frac{x^{2/3}}{3x^{1/3} + x + x^2}$$

Factoring out in the denominator the power with the minimum exponent, we have:

$$\begin{aligned} \frac{x^{2/3}}{3x^{1/3} + x + x^2} &= \frac{x^{2/3}}{3x^{1/3} \left(1 + \frac{x}{3x^{1/3}} + \frac{x^2}{3x^{1/3}} \right)} \\ &= \frac{x^{2/3}}{3x^{1/3} \left(1 + \frac{x^{2/3}}{3} + \frac{x^{5/3}}{3} \right)} \\ &= \frac{1}{3} x^{1/3} \frac{1}{\left(1 + \frac{x^{2/3}}{3} + \frac{x^{5/3}}{3} \right)} \end{aligned}$$

Example: powers with rational exponents (2/2)

Hence:

$$\lim_{x \rightarrow 0} \frac{x^{2/3}}{3x^{1/3} + x + x^2} = \frac{1}{3} \lim_{x \rightarrow 0} x^{1/3} \frac{1}{\left(1 + \underbrace{\frac{x^{2/3}}{3} + \frac{x^{5/3}}{3}}_{\rightarrow 0 \text{ as } x \rightarrow 0}\right)} = 0$$

Change of variable

Another option to compute limits as $x \rightarrow 0$ is to make a change of variable:

$$x = \frac{1}{y},$$

if $x \rightarrow 0^+$ then $y \rightarrow +\infty$, if $x \rightarrow 0^-$ then $y \rightarrow -\infty$

and hence we reduce to the case of limits at $\pm\infty$.

Example: change of variable as $x \rightarrow 0$ (1/4)

We want to compute the following limit:

$$\lim_{x \rightarrow 0} \frac{x^{2/3}}{3x^{1/3} + x + x^2}, \quad f(x) = \frac{x^{2/3}}{3x^{1/3} + x + x^2}$$

Making the change of variable $x = \frac{1}{y}$, we therefore have to compute the following two limits:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{y \rightarrow +\infty} \frac{\left(\frac{1}{y}\right)^{2/3}}{3\left(\frac{1}{y}\right)^{1/3} + \frac{1}{y} + \left(\frac{1}{y}\right)^2}$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{y \rightarrow -\infty} \frac{\left(\frac{1}{y}\right)^{2/3}}{3\left(\frac{1}{y}\right)^{1/3} + \frac{1}{y} + \left(\frac{1}{y}\right)^2}$$

Example: change of variable as $x \rightarrow 0$ (2/4)

We have:

$$\frac{\left(\frac{1}{y}\right)^{2/3}}{3 \left(\frac{1}{y}\right)^{1/3} + \frac{1}{y} + \left(\frac{1}{y}\right)^2} = \frac{y^{-2/3}}{3 y^{-1/3} + y^{-1} + y^{-2}}$$

Factoring out in the denominator the power with the maximum exponent, we have:

$$\begin{aligned} \frac{y^{-2/3}}{3 y^{-1/3} + y^{-1} + y^{-2}} &= \frac{y^{-2/3}}{3 y^{-1/3} \left(1 + \frac{y^{-1}}{3 y^{-1/3}} + \frac{y^{-2}}{3 y^{-1/3}}\right)} \\ &= \frac{1}{3} \frac{1}{y^{1/3}} \frac{1}{\left(1 + \frac{1}{3 y^{2/3}} + \frac{1}{3 y^{5/3}}\right)} \end{aligned}$$

Example: change of variable as $x \rightarrow 0$ (3/4)

Hence:

$$\begin{aligned} & \frac{1}{3} \lim_{y \rightarrow -\infty} \frac{1}{y^{1/3}} \frac{1}{\left(1 + \frac{1}{3y^{2/3}} + \frac{1}{3y^{5/3}}\right)} \\ &= \frac{1}{3} \lim_{y \rightarrow +\infty} \frac{1}{y^{1/3}} \frac{1}{\left(1 + \frac{1}{3y^{2/3}} + \frac{1}{3y^{5/3}}\right)} = 0 \end{aligned}$$

since, as $y \rightarrow \pm\infty$, we have

$$\frac{1}{y^{1/3}} \rightarrow 0 \quad \text{and} \quad \left(1 + \frac{1}{3y^{2/3}} + \frac{1}{3y^{5/3}}\right) \rightarrow 1$$

because $\frac{1}{3y^{2/3}} \rightarrow 0$ and $\frac{1}{3y^{5/3}} \rightarrow 0$.

Example: change of variable as $x \rightarrow 0$ (4/4)

Since the right limit and the left limit of $f(x)$ as $x \rightarrow 0$ exist and are equal to 0, then:

$$\lim_{x \rightarrow 0} f(x) = 0$$

Summary

- ▶ For quotients of sums of powers with rational exponents: as $x \rightarrow \infty$ we factor out the power with the maximum exponent, as $x \rightarrow 0^+$ the one with the minimum exponent
- ▶ With rational exponents having an odd denominator the base can be negative, so we can consider $x \rightarrow 0$
- ▶ The change of variable $x = \frac{1}{y}$ turns $x \rightarrow 0^\pm$ into $y \rightarrow \pm\infty$

Final summary

Limits at $\pm\infty$:

- ▶ Polynomials: only the monomial of maximum degree matters
- ▶ Rational functions: the ratio of the monomials of maximum degree matters, $\frac{a_n}{b_m} x^{n-m}$

Limits as $x \rightarrow 0$:

- ▶ We factor out the powers of minimum degree (exponent)
- ▶ Alternatively we use the change of variable $x = \frac{1}{y}$

Powers with rational exponents:

- ▶ The same rules hold; with an odd denominator we can study $x \rightarrow 0$